CODE REVIEW OF THE PROJECT

Senior & Junior

1. **Airfoil Screening & Ranking Code: PYTHON (**rank_38_glider_airfoils.py **)**

- **Inputs & Justifications**

★ **Input Data:** Raw 2D XFLR5 polar files containing [α, C_l, C_d, C_m] sweeps.

★ **Operating Range:** Low Reynolds number ($Re \approx 25,000–30,000$).

★ **Justification:** At glider scales ($V \approx 6.0 m/s$), viscous laminar separation dominates, requiring direct low-Re polar evaluation to avoid stall-prone profiles.

- **Workflow & Code Logic**

★ **Batch Ingestion:** Automatically reads and parses header metadata (name, Re) and numeric records.

★ **Feature Extraction:** Tracks peak glide $(C_l/C_d)_{\max}$, power factor $(C_l^{1.5}/C_d)_{\max}$, and trim moment C_{m0}.

★ **Min-Max Normalization:** Scales metrics from 0.0 to 1.0 so parameters with different units have equal weight.

★ **Multi-Attribute Utility Function:**

$$Score = \left[0.50 \times \overline{\left(\frac{C_l}{C_d}\right)} + 0.35 \times \overline{\left(\frac{C_l^{1.5}}{C_d}\right)} + 0.15 \times \left(1 - \overline{|C_{m0}|}\right)\right] \times 100$$

(50% Glide Range + 35% Hang Time + 15% Pitch Trim Penalty).

- **Intermediate Outputs**

★ Peak glide ratio $(C_l/C_d)_{\max}$ and its design angle α_{opt}.

★ Minimum sink endurance metric $(C_l^{1.5}/C_d)_{\max}$.

★ Zero-lift pitching moment penalty C_{m0} at $\alpha \approx 0^{\circ}$.

- **Final Design Outputs**
  - ★ Filtered **Top 10 Leaderboard** printed to the console.
  - ★ Exported shortlists: ALL_RANKED_AIRFOILS.csv and TOP_10_WEIGHTED_AIRFOILS.csv.
  - ★ Winning sectional baseline profile and its design cruise lift coefficient ($C_{L,cruise}$).

- **Stage Coupling (How it feeds the next stage)**
  - ★ Passes the chosen candidate's $C_{L,cruise}$ and α_{cruise} into the 2D-to-3D finite wing conversion / wing sizing stage to calculate reference area (S_w), mean chord ($\bar{c}$), and wing incidence (i_w).
  - ★ Passes the airfoil C_{m0} forward to the tail design stage to set the required trim decalage ($\Delta i = i_w - i_t$).

2. MATLAB Airfoil Ranking & Multi-Attribute Pipeline ([top_10_rank_glider_airfoils.m])

It is the direct MATLAB equivalent of the Python screening code, sharing the exact same mathematical scoring logic, deduplication principle, and weighting criteria.

Key Enhancement: Along with terminal ranking and CSV export, this MATLAB script automatically plots a 4-quadrant comparative figure ($C_l vs C_d$, $C_l vs \alpha$, $C_l/C_d vs \alpha$, $C_m vs \alpha$) for the top 5 contenders.

- **Inputs & Justifications**
  - ★ Input Data: Tabular ASCII polar files (.txt, .plr, .csv) inside polar_data/ exported from XFLR5.
  - ★ Header Cutoff: Reads numeric data with readmatrix('HeaderLines', 11) to cleanly bypass text metadata.
  - ★ Justification: Automates parsing across 38+ airfoils without manual plotting bias at operational low Reynolds numbers ($Re \approx 25,000–30,000$).

- **Workflow & Code Logic**
  - ★ Metadata & Data Parsing: Extracts airfoil name and Reynolds number from header lines; reads $[\alpha, C_l, C_d, C_m]$ arrays.
  - ★ Filtering: Retains only positive physical drag/lift points (cd > 0 & cl > 0).
  - ★ Metric Extraction: Computes L/D and $(C_l^{1.5}/C_d)$; extracts peak values and zero-lift pitching moment (C_{m0}).
  - ★ Deduplication: Keeps only the single highest-performing run per airfoil family using a MATLAB containers.Map.
  - ★ Normalized Scoring: Applies Min-Max scaling $(0.0 to 1.0)$ and evaluates:

$$Score = \left[0.50 \times \overline{\left(\frac{C_l}{C_d}\right)} + 0.35 \times \overline{\left(\frac{C_l^{1.5}}{C_d}\right)} + 0.15 \times \left(1 - \overline{|C_{m0}|}\right)\right] \times 100$$

(50% Glide Distance + 35% Flight Time + 15% Pitch Trim Penalty).

  - ★ Visualization: Generates 4-quadrant comparison curves for the Top 5 candidates.

- **Intermediate Outputs**
  - ★ Peak glide ratio $(C_l/C_d)_{\max}$ and optimum cruise angle α_{opt}.
  - ★ Maximum endurance parameter $(C_l^{1.5}/C_d)_{\max}$.
  - ★ Trim pitching moment C_{m0} near $\alpha = 0^{\circ}$.

- **Final Design Outputs**
  - ★ Sorted Top 10 Leaderboard printed to the MATLAB Command Window.
  - ★ Exported CSV file: TOP_10_WEIGHTED_AIRFOILS.csv.
  - ★ Consolidated 4-Quadrant Comparative Figure validating the top 5 airfoils.

- **Stage Coupling (Feeding the Next Design Stage)**
  - ★ The top 10 profiles identified by this script directly advance to 3D Finite-Wing Verification (flow5 / VLM2) to account for induced drag ($C_{Di} = \frac{C_L^2}{\pi e AR}$) and downwash.
  - ★ The winning profile's design lift coefficient ($C_{L,cruise}$) and angle of attack are used downstream to compute the physical wing area (S_w), mean chord ($\bar{c}$), and wing mounting angle (i_w).

3. 3D Finite-Wing & Glider Configuration Ranking (3D_ranking_of_top_10.m)

Evaluates full **3D finite-wing aircraft polars** exported from flow5 / XFLR5 (VLM2 / 3D panel analysis).

Moves beyond 2D infinite-wing theory by accounting for real physical losses: **induced drag** ($C_{Di} = \frac{C_L^2}{\pi e AR}$), wingtip vortices, downwash ($\alpha_{eff} = \alpha - \alpha_i$), and complete aircraft pitch equilibrium (C_m).

★ **Inputs & Justifications**

- ★ **Input Data:** Multi-row text files inside Best3DT2_polars/ generated from flow5 Type 2 (fixed-lift, $m \approx 35g$, $V \approx 6.0 m/s$) sweeps.
- ★ **Parser Logic:** Reads wrapped flow5 data blocks by detecting lines starting with token '0' (control state flag).
- ★ **Extracted Columns:** Token 2 (α), Token 5 (C_L), Token 6 (C_D), and Token 10 (C_m).
- ★ **Data Filter:** cd_val > 0.005 to ignore unphysical solver anomalies or numerical divergence.

★ Workflow & Code Logic

- ★ **Block-by-Block Extraction:** Parses multiline text records and extracts true 3D aerodynamic coefficients.
- ★ **Cruise Condition Capture:** Locates peak glide ratio $(C_L/C_D)_{\max}$ and extracts the exact operating lift ($C_{L,cruise}$) and drag ($C_{D,cruise}$) at that condition.
- ★ **Endurance & Trim:** Tracks maximum 3D loiter factor ($C_L^{1.5}/C_D$) and zero-lift pitch moment (C_{m0}).
- ★ **Min-Max Feature Scaling:** Scales metrics to a $0.0 to 1.0$ baseline.
- ★ **Composite Utility Function:**

$$Score = \left[0.50 \times \overline{\left(\frac{C_l}{C_d}\right)} + 0.35 \times \overline{\left(\frac{C_l^{1.5}}{C_d}\right)} + 0.15 \times \left(1 - \overline{|C_{m0}|}\right)\right] \times 100$$

(50% 3D Ground Distance + 35% Flight Hang Time + 15% Pitch Trim Penalty).

★ Intermediate Outputs

- ★ Peak 3D finite glide ratio $(C_L/C_D)_{\max}$ (typically drops from 2D values of ~30 down to ~12–14 due to induced drag).
- ★ Operating cruise lift coefficient $C_{L@\max L/D}$ and total drag $C_{D@\max L/D}$.
- ★ 3D loiter factor and pitch trim moment C_{m0}.

★ Final Design Outputs

- ★ Sorted **3D Configuration Leaderboard** printed to the console.
- ★ Shortlist export: TOP_10_3D_RANKED_GLIDERS.csv.
- ★ Identification of the **winning airfoil profile** and design operating point for the physical glider.

★ Stage Coupling (Feeding Downstream Design Steps)

- ★ **Main Wing Geometry:** The confirmed 3D cruise lift coefficient ($C_{L@\max L/D}$) directly sizes the wing reference area ($S_w = \frac{2W}{\rho V^2 C_L}$) and root/tip chords ($c_r, c_t$).

★ **Empennage Sizing:** The 3D pitching moment (C_m) and wing lift curve slope feed directly into horizontal stabilizer sizing (S_t, l_t) to establish the neutral point (x_{NP}) and ensure a positive static margin (SM ≈10%).

4. Parametric Aspect Ratio Sweep & Final Configuration Ranking (FINAL_aerofoil_rank_with_AR.m)

Evaluates the **Top 3 candidate airfoils** across an **Aspect Ratio parametric sweep** ($AR = 5.0\ to\ 7.2$).

Resolves the engineering trade-off: higher AR slashes lift-induced drag ($C_{Di} = \frac{C_L^2}{\pi e AR}$), but shrinks the physical chord ($\bar{c} = b/AR$), driving the local Reynolds number down and risking premature laminar separation.

- **Inputs & Justifications**
  - ★ **Input Data:** 3D VLM2 polars inside Best_03_analysis/ generated from flow5 fixed-lift simulations ($W = 0.343N, V = 6.0m/s$).
  - ★ **Filename Parsing:** Uses regex to automatically decouple the airfoil name and numeric Aspect Ratio from filenames (e.g., AG12_5.4, AG14_AR6.5).
  - ★ **Filtered Columns:** Scans flow5 wrapped records via the '0' control flag for α, C_L, C_D, and pitching moment C_m.

- **Workflow & Code Logic**
  - ★ **Batch Parameter Extraction:** Iterates through all sweep files and pairs each run with its respective airfoil and Aspect Ratio.
  - ★ **Performance Metrics:** Tracks peak 3D glide $(C_L/C_D)_{\max}$, optimal angle α_{opt}, cruise lift $C_{L@\max L/D}$, cruise drag $C_{D@\max L/D}$, and C_{m0}.
  - ★ **Multi-Attribute Utility Function:**

$$Score = \left[0.50 \times \overline{\left(\frac{C_l}{C_d}\right)} + 0.35 \times \overline{\left(\frac{C_l^{1.5}}{C_d}\right)} + 0.15 \times \left(1 - \overline{|C_{m0}|}\right)\right] \times 100$$

(50% 3D Ground Distance + 35% Flight Hang Time + 15% Pitch Trim Penalty).

★ **Sensitivity Visualization:** Plots $(C_L/C_D)_{\max}$ vs. AR curves for all top contenders to pinpoint the optimal Aspect Ratio peak.

- **Intermediate Outputs**
  - ★ Direct variation of $(C_L/C_D)_{\max}$ across different AR values for each finalist.
  - ★ Extracted operational cruise lift (C_L) and drag (C_D) for every tested geometry.
  - ★ Pitch trim stability offsets (C_{m0}).

- **Final Design Outputs**
  - ★ **Leaderboard & Overall Verdict:** Identifies the single winning configuration, profile, and optimal Aspect Ratio.
  - ★ **Exported Dataset:** PARAMETRIC_AR_RANKING_RESULTS.csv.
  - ★ **Sensitivity Curve:** A plot of 3D Glide Ratio vs. Wing Aspect Ratio clearly showing the efficiency plateau.

- **Stage Coupling (Feeding the Next Design Stage)**
  - ★ **Freezes Main Wing Sizing:** The winning airfoil and optimal AR set the reference wing area (S_w), wingspan ($b = 400mm$), root chord (c_r), tip chord (c_t), and MAC ($\bar{c}$).
  - ★ **Feeds Tail Sizing & Placement:** The frozen wing area (S_w) and mean chord ($\bar{c}$) pass directly into horizontal stabilizer sizing via the tail volume coefficient ($V_h \approx 0.50$) and set the tail moment arm ($l_t \approx 3\bar{c}$).

5. Tail Airfoil Screening & Ranking Code: (Tail_aerofoil_ranking.m)

- **Inputs & Justifications**
  - ★ **Input Data:** Raw 2D XFLR5 polar files (.txt, .plr, .csv) containing $[\alpha, C_l, C_d, C_m]$ sweeps for symmetric tail airfoils.
  - ★ **Operating Range:** Low Reynolds number regime, dynamically parsed from the polar file headers.
  - ★ **Justification:** Tail surfaces predominantly operate near zero angle of attack ($\alpha \approx 0^{\circ}$). Evaluating control deadband (responsiveness) and baseline parasite drag is critical for stability without excessive drag penalties.

- **Workflow & Code Logic**
  - ★ **Batch Ingestion:** Automatically unzips, reads, and parses header metadata (airfoil name, Re) and numeric records for all tested tail airfoils.
  - ★ **Feature Extraction:** Tracks Zero-lift parasite drag (C_{d0}) at $\alpha \approx 0^{\circ}$ and computes the Lift curve slope ($dC_l/d\alpha$) linearly between -2° and $+2^{\circ}$.
  - ★ **Min-Max Normalization:** Scales the slope and drag metrics from 0.0 to 1.0 so parameters with different units have equal weight.
  - ★ **Multi-Attribute Utility Function:**

$$Score = \left[0.50 \times \overline{\left(\frac{dC_l}{d\alpha}\right)} + 0.50 \times \left(1 - \overline{C_{d0}}\right)\right] \times 100$$

(50% Control Authority / Slope + 50% Low Drag Floor).

- **Intermediate Outputs**
  - ★ Lift curve slope ($dC_l/d\alpha$) near neutral pitch to identify severe deadbands.
  - ★ Zero-lift parasite drag metric (C_{d0}).
  - ★ Categorization of the slope into a "Deadband Rating" (ranging from Excellent (Sharp) to Poor (Severe Deadband)).

- **Final Design Outputs**

★ Filtered Complete Leaderboard printed to the MATLAB console with terminal scores and Deadband Ratings.
★ Exported ranking data: TAIL_AIRFOILS_RANKED.csv.
★ Winning symmetric tail baseline profile.
★ Consolidated 2-Quadrant Comparative Figure visually validating the lift curve and parasite drag floor of the candidates.

- **Stage Coupling (How it feeds the next stage)**
  - ★ Passes the chosen optimal symmetric profile forward to the Tail Dimension Analysis stage.
  - ★ Ensures the final full-aircraft 3D VLM simulation utilizes a highly responsive tail section that minimizes zero-lift drag contribution while maintaining the required Static Margin (SM).

6. Tail Dimension Analysis & DFM Ranking Code: MATLAB (tail_dimension_ranking.m)

- **Inputs & Justifications**
  - ★ **Input Data:** 3D polar text files containing parametric tail sweeps, combined with the frozen main wing reference area ($S_w = 0.024m^2$) and mean aerodynamic chord ($c_{mac} = 0.060m$).
  - ★ **Operating Range:** Extracts geometric parameters (Tail Area S_t, Aspect Ratio AR_t, Moment Arm l_t) directly from configuration filenames.
  - ★ **Justification:** Tail sizing must balance aerodynamic efficiency, longitudinal stability (pitch stiffness), and Design-for-Manufacturing (DFM) practicality. A strictly aerodynamic tail might be too thin to fabricate, requiring a dedicated DFM scoring approach.

- **Workflow & Code Logic**
  - ★ **Batch Ingestion:** Unzips and parses the parametric tail polar files, mathematically resolving physical dimensions like tail span (b_t) and tail chord (c_t).
  - ★ **Feature Extraction:** Tracks aerodynamic metrics (Peak L/D, Hang time $C_L^{1.5}/C_D$) and the critical pitch stiffness slope ($dC_m/d\alpha$).
  - ★ **Multi-Attribute Utility Function (DFM Weighted):**

$$Score = (0.45 \times L/D) + (0.25 \times Endurance) + (0.15 \times Stiffness) + (0.15 \times Chord_{DFM})$$

(Rewards wider chords to ensure structural buildability and better low-Reynolds health).

  - ★ **Stability Gatekeeper:** Dynamically calculates the Tail Volume Coefficient (V_h). If V_h falls outside the stable optimal range of $0.40 - 0.60$, the code applies a severe 35% score penalty to instantly demote unstable configurations.

- **Intermediate Outputs**
  - ★ Calculated physical build dimensions: Tail span (b_t) and chord (c_t) in millimeters.
  - ★ Computed Tail Volume Coefficient (V_h) verifying the neutral point placement.
  - ★ Pitch moment slope ($C_{m,\alpha}$) validating longitudinal static stability.

- **Final Design Outputs**
  - ★ DFM-Optimized Leaderboard printed to the console displaying physical dimensions, stability status, and terminal scores.
  - ★ Exported design matrix: TAIL_ANALYSIS_DFM_RANKED.csv.
  - ★ The definitive winning tail configuration outputting exact physical dimensions (Span, Chord, Moment Arm) required for fabrication.

- **Stage Coupling (How it feeds the next stage)**
  - ★ Freezes the complete horizontal stabilizer geometry (S_t, b_t, c_t, l_t) and computationally locks in the longitudinal static margin.

★ Passes these finalized physical dimensions directly forward to the Manufacturing Stage and safely opens the gateway to begin the Vertical Stabilizer analysis.